Longest Dist (LD)

LD (0, 2) = 2

LD (2, 1) = 2

LD (0, 1) = LD(0,2) + LD(2,1)

2 = 2 + 2

S1 = ABCDGM

S2 = AEBFMR

LCS(A B C D G M, A E B F M R) =

= 1 + LCS(B C D G M, E B F M R)

LCS(C D G M, E B F M R) LCS(B C D G M, B F M R)

LCS(i, j) = 1 + LCS(i+1, j+1) if s1[i] == s2[j]

= max(LCS(i+1, j), LCS(i, j+1)) otherwise.

TC = $O(2^{max(n,m)})$.